do-klasteryzacji.R

Jamistrz

2024-06-22

```r
anskup <- read.csv("segmentation.csv")
anskup$ID = NULL
View(anskup)
attach(anskup)
library(lmtest)
```

```
## Ładowanie wymaganego pakietu: zoo
```

```
##
## Dołączanie pakietu: 'zoo'
```

```
## Następujące obiekty zostały zakryte z 'package:base':
##
##     as.Date, as.Date.numeric
```

```r
library(car)
```

```
## Ładowanie wymaganego pakietu: carData
```

```r
library(corrplot)
```

```
## corrplot 0.92 loaded
```

```r
library(cluster)
library(fpc)
Y1<-data.frame(anskup)
k<-ncol(Y1)
k
```

```
## [1] 7
```

```r
head(Y1)
```

```
##   Sex Marital.status Age Education Income Occupation Settlement.size
## 1   0              0  67         2 124670          1               2
## 2   1              1  22         1 150773          1               2
## 3   0              0  49         1  89210          0               0
## 4   0              0  45         1 171565          1               1
## 5   0              0  53         1 149031          1               1
## 6   0              0  35         1 144848          0               0
```

```r
#standarization
K <- Y1
K[, c(3,4,5,6,7)] <- scale(Y1[, c(3,4,5,6,7)])
K1 <- data.frame(Sex, K[, -1])
K1 = data.frame(id=1:20, K1)
head(K1)
```

```
##   id Sex Marital.status        Age  Education      Income Occupation
## 1  1   0              0  1.8807195  1.6289408 -0.39588552  0.1560736
## 2  2   1              1 -1.2080534 -0.1809934  0.23403985  0.1560736
## 3  3   0              0  0.6452103 -0.1809934 -1.25161680 -1.4046626
## 4  4   0              0  0.3706527 -0.1809934  0.73579859  0.1560736
## 5  5   0              0  0.9197679 -0.1809934  0.19200139  0.1560736
## 6  6   0              0 -0.3157412 -0.1809934  0.09105599 -1.4046626
##   Settlement.size
## 1       1.4396969
## 2       1.4396969
## 3      -0.9597980
## 4       0.2399495
## 5       0.2399495
## 6      -0.9597980
```

```r
"dendogram"
```

```
## [1] "dendogram"
```

```r
#method - "euclidean", "maximum", "manhattan", "canberra", "binary" or "minkowski"
"tworzymy macierz odleglosci"
```

```
## [1] "tworzymy macierz odleglosci"
```

```r
dist.K<-dist(K1[2:k],method = "euclidian",diag = TRUE)
round(dist.K,2)
```

```
##       1    2    3    4    5    6    7    8    9   10   11   12   13   14   15
## 1  0.00
## 2  3.90 0.00
## 3  2.82 3.17 0.00
## 4  2.61 2.18 2.54 0.00
## 5  2.13 2.56 2.14 0.77 0.00
## 6  3.28 2.29 1.65 1.82 1.99 0.00
## 7  2.19 2.56 2.27 0.66 0.18 2.01 0.00
## 8  3.65 2.51 4.13 1.79 2.26 3.34 2.18 0.00
## 9  2.01 3.73 2.69 2.85 2.65 2.74 2.65 4.27 0.00
## 10 3.92 1.98 4.13 2.19 2.60 3.39 2.56 1.20 4.30 0.00
## 11 3.71 1.04 2.71 2.49 2.58 2.38 2.65 3.02 3.73 2.46 0.00
## 12 3.74 0.58 2.87 2.28 2.50 2.28 2.54 2.75 3.67 2.19 0.47 0.00
## 13 3.87 1.02 2.84 2.61 2.73 2.45 2.80 3.07 3.87 2.48 0.21 0.48 0.00
## 14 1.84 4.09 1.96 3.27 2.83 2.83 2.93 4.72 1.81 4.80 3.70 3.83 3.84 0.00
## 15 3.57 1.23 2.57 2.47 2.49 2.39 2.57 3.08 3.64 2.54 0.25 0.66 0.43 3.56 0.00
## 16 3.77 2.27 1.83 3.04 2.94 1.97 3.04 4.27 3.25 3.89 1.70 1.91 1.77 3.00 1.62
## 17 2.05 2.63 1.81 1.24 0.57 2.04 0.75 2.58 2.72 2.79 2.42 2.44 2.59 2.59 2.29
## 18 4.69 2.31 3.34 2.30 2.65 2.51 2.64 2.64 4.76 2.63 2.58 2.41 2.59 4.90 2.66
## 19 4.23 3.90 5.70 3.53 3.94 4.84 3.82 2.89 4.38 3.15 4.60 4.26 4.66 5.50 4.65
## 20 4.41 2.38 3.00 2.15 2.36 2.41 2.37 2.68 4.56 2.70 2.49 2.38 2.53 4.61 2.52
##      16   17   18   19   20
## 1
## 2
## 3
## 4
## 5
## 6
## 7
## 8
## 9
## 10
## 11
## 12
## 13
## 14
## 15
## 16 0.00
## 17 2.70 0.00
## 18 3.27 2.75 0.00
## 19 5.59 4.35 5.06 0.00
## 20 3.09 2.40 0.51 5.13 0.00
```

```r
#minimum of distance
min(dist.K)
```

```
## [1] 0.1801235
```

```
#hierarchical grouping - "ward.D2", "single", "complete", "average" (= UPGMA), "mcquitty" (= WPGMA), "median" (= WPGMC) or
"centroid" (= UPGMC).
CW<-hclust(dist.K,method = "ward.D2",members = NULL)
"polaczenia"
```

```
## [1] "polaczenia"
```

```
CW$merge
```

```
##       [,1] [,2]
##  [1,]   -5   -7
##  [2,]  -11  -13
##  [3,]  -15    2
##  [4,]  -18  -20
##  [5,]   -2  -12
##  [6,]  -17    1
##  [7,]   -4    6
##  [8,]   -8  -10
##  [9,]    3    5
## [10,]   -3   -6
## [11,]   -9  -14
## [12,]   -1   11
## [13,]  -16   10
## [14,]  -19    8
## [15,]    7   13
## [16,]    4    9
## [17,]   12   15
## [18,]   14   16
## [19,]   17   18
```

```
"rosnaco odleglocci"
```

```
## [1] "rosnaco odleglocci"
```

```
CW$height
```

```
##  [1] 0.1801235 0.2059854 0.3901917 0.5144723 0.5759149 0.7583060 1.0634647
##  [8] 1.2002245 1.2334853 1.6511203 1.8052786 1.9647441 1.9824307 3.4231893
## [15] 3.9103574 4.1115340 4.5301185 5.8713446 7.2680488
```

```
"uporzadkowania obiektoW"
```

```
## [1] "uporzadkowania obiektoW"
```

```
CW$order
```

```
##  [1]  1  9 14  4 17  5  7 16  3  6 19  8 10 18 20 15 11 13  2 12
```

```
"wybrana metoda aglomeracji"
```

```
## [1] "wybrana metoda aglomeracji"
```

```
CW$method
```

```
## [1] "ward.D2"
```

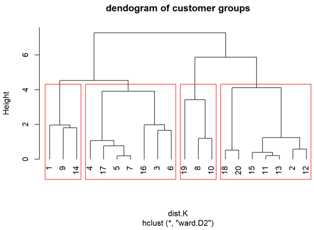
```
"Tworzenie dendogramu"
```

```
## [1] "Tworzenie dendogramu"
```

```
A1<-plot(CW,labels = K1[,1],main="dendogram of customer groups",hang=-5)
"ustawiajac k=ilosc klastrow mozemy dostac podzial"
```

```
## [1] "ustawiajac k=ilosc klastrow mozemy dostac podzial"
```

```
rect.hclust(CW,k=4,border = "red")
```



```
"tworzenie klastrow z przypisanymi elementami"
```

```
## [1] "tworzenie klastrow z przypisanymi elementami"
```

```
person.clust3<-cutree(CW,k=4)
person.clust3
```

```
##  [1] 1 2 3 3 3 3 3 4 1 4 2 2 2 1 2 3 3 2 4 2
```

```
"instrukcie laczace klastry z osobami"
```

```
## [1] "instrukcie laczace klastry z osobami"
```

```
A<-data.frame(person.clust3)
A
```

```
##    person.clust3
## 1              1
## 2              2
## 3              3
## 4              3
## 5              3
## 6              3
## 7              3
## 8              4
## 9              1
## 10             4
## 11             2
## 12             2
## 13             2
## 14             1
## 15             2
## 16             3
## 17             3
## 18             2
## 19             4
## 20             2
```

```
B<-cbind(K1[,1],A)
B
```

```
##    K1[, 1] person.clust3
## 1        1             1
## 2        2             2
## 3        3             3
## 4        4             3
## 5        5             3
## 6        6             3
## 7        7             3
## 8        8             4
## 9        9             1
## 10      10             4
## 11      11             2
## 12      12             2
## 13      13             2
## 14      14             1
## 15      15             2
## 16      16             3
## 17      17             3
## 18      18             2
## 19      19             4
## 20      20             2
```

```
"szczegoly analiz z podzialem na 3 klastry wartosci
zmiennych dla kazdego elementu w klastrze"
```

```
## [1] "szczegoly analiz z podzialem na 3 klastry wartosci \nzmiennych dla kazdego elementu w klastrze"
```

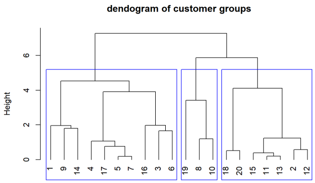
```
person.clust<-lapply(1:4,function(which.group)K[person.clust3==which.group,])
person.clust
```

```
## [[1]]
##    Sex Marital.status      Age Education     Income Occupation Settlement.size
## 1    0              0 1.880719  1.628941 -0.3958855  0.1560736        1.439697
## 9    0              1 1.468883  1.628941  0.2537801 -1.4046626       -0.959798
## 14   0              0 1.400244  1.628941 -1.2476591 -1.4046626       -0.959798
## 
## [[2]]
##    Sex Marital.status        Age  Education      Income Occupation
## 2    1              1 -1.2080534 -0.1809934  0.23403985  0.1560736
## 11   1              1 -1.0021352 -0.1809934 -0.78685287  0.1560736
## 12   1              1 -1.0707746 -0.1809934 -0.32527442  0.1560736
## 13   1              1 -1.2080534 -0.1809934 -0.78159203  0.1560736
## 15   1              1 -0.7962170 -0.1809934 -0.92126977  0.1560736
## 18   0              0 -1.0021352 -1.9909277  0.39478509  0.1560736
## 20   0              0 -0.5902988 -1.9909277  0.08644672  0.1560736
##    Settlement.size
## 2        1.4396969
## 11      -0.9597980
## 12      -0.9597980
## 13       1.4396969
## 15       0.2399495
## 18       1.4396969
## 20       0.2399495
## 
## [[3]]
##    Sex Marital.status        Age  Education      Income Occupation
## 3    0              0  0.6452103 -0.1809934 -1.25161680 -1.4046626
## 4    0              0  0.3706527 -0.1809934  0.73579859  0.1560736
## 5    0              0  0.9197679 -0.1809934  0.19200139  0.1560736
## 6    0              0 -0.3157412 -0.1809934  0.09105599 -1.4046626
## 7    0              0  0.9197679 -0.1809934  0.37212486  0.1560736
## 16   1              1 -0.5216594 -0.1809934 -1.27048825 -1.4046626
## 17   0              0  0.9197679 -0.1809934 -0.37464910  0.1560736
##    Settlement.size
## 3       -0.9597980
## 4        0.2399495
## 5        0.2399495
## 6       -0.9597980
## 7        0.2399495
## 16      -0.9597980
## 17      -0.9597980
## 
## [[4]]
##    Sex Marital.status        Age  Education    Income Occupation
## 8    0              0 -0.3157412 -0.1809934 1.2680605    1.71681
## 10   0              1 -0.7962170 -0.1809934 0.8101502    1.71681
## 19   1              1  0.3020133  1.6289408 2.9170446    1.71681
##    Settlement.size
## 8        0.2399495
## 10      -0.9597980
## 19       1.4396969
```

```
"analogiczne rozwa?ania w podziale na dwa klastry"
```

```
## [1] "analogiczne rozwa?ania w podziale na dwa klastry"
```

```
A2<-plot(CW,labels = K1[,1],main="dendogram of customer groups",hang=-1)
rect.hclust(CW,k=3,border = "blue")
```



dist.K
hclust (*, "ward.D2")

```
person.clust2<-cutree(CW,k=3)
person.clust2
```

```
##  [1] 1 2 1 1 1 1 1 3 1 3 2 2 2 1 2 1 1 2 3 2
```

```
A3<-data.frame(person.clust2)
A3
```

```
##    person.clust2
## 1              1
## 2              2
## 3              1
## 4              1
## 5              1
## 6              1
## 7              1
## 8              3
## 9              1
## 10             3
## 11             2
## 12             2
## 13             2
## 14             1
## 15             2
## 16             1
## 17             1
## 18             2
## 19             3
## 20             2
```

```
B1<-cbind(K1[,1],A3)
B1
```

```
##    K1[, 1] person.clust2
## 1        1             1
## 2        2             2
## 3        3             1
## 4        4             1
## 5        5             1
## 6        6             1
## 7        7             1
## 8        8             3
## 9        9             1
## 10      10             3
## 11      11             2
## 12      12             2
## 13      13             2
## 14      14             1
## 15      15             2
## 16      16             1
## 17      17             1
## 18      18             2
## 19      19             3
## 20      20             2
```

```
person.clust22<-lapply(1:3,function(which.group)K[person.clust2==which.group,])
person.clust22
```

```
## [[1]]
##    Sex Marital.status        Age  Education      Income Occupation
## 1    0              0  1.8807195  1.6289408 -0.39588552  0.1560736
## 3    0              0  0.6452103 -0.1809934 -1.25161680 -1.4046626
## 4    0              0  0.3706527 -0.1809934  0.73579859  0.1560736
## 5    0              0  0.9197679 -0.1809934  0.19200139  0.1560736
## 6    0              0 -0.3157412 -0.1809934  0.09105599 -1.4046626
## 7    0              0  0.9197679 -0.1809934  0.37212486  0.1560736
## 9    0              1  1.4688831  1.6289408  0.25378007 -1.4046626
## 14   0              0  1.4002437  1.6289408 -1.24765910 -1.4046626
## 16   1              1 -0.5216594 -0.1809934 -1.27048825 -1.4046626
## 17   0              0  0.9197679 -0.1809934 -0.37464910  0.1560736
##    Settlement.size
## 1        1.4396969
## 3       -0.9597980
## 4        0.2399495
## 5        0.2399495
## 6       -0.9597980
## 7        0.2399495
## 9       -0.9597980
## 14      -0.9597980
## 16      -0.9597980
## 17      -0.9597980
## 
## [[2]]
##    Sex Marital.status        Age  Education      Income Occupation
## 2    1              1 -1.2080534 -0.1809934  0.23403985  0.1560736
## 11   1              1 -1.0021352 -0.1809934 -0.78685287  0.1560736
## 12   1              1 -1.0707746 -0.1809934 -0.32527442  0.1560736
## 13   1              1 -1.2080534 -0.1809934 -0.78159203  0.1560736
## 15   1              1 -0.7962170 -0.1809934 -0.92126977  0.1560736
## 18   0              0 -1.0021352 -1.9909277  0.39478509  0.1560736
## 20   0              0 -0.5902988 -1.9909277  0.08644672  0.1560736
##    Settlement.size
## 2        1.4396969
## 11      -0.9597980
## 12      -0.9597980
## 13       1.4396969
## 15       0.2399495
## 18       1.4396969
## 20       0.2399495
## 
## [[3]]
##    Sex Marital.status        Age  Education    Income Occupation
## 8    0              0 -0.3157412 -0.1809934 1.2680605    1.71681
## 10   0              1 -0.7962170 -0.1809934 0.8101502    1.71681
## 19   1              1  0.3020133  1.6289408 2.9170446    1.71681
##    Settlement.size
## 8        0.2399495
## 10      -0.9597980
## 19       1.4396969
```

```
"comparative statistics of clusters -
r??ne statystyki dla klastr?w"
```

```
## [1] "comparative statistics of clusters - \nr??ne statystyki dla klastr?w"
```

```
R<-cluster.stats(dist.K,person.clust3,person.clust2)
R
```

```
## $n
## [1] 20
##
## $cluster.number
## [1] 4
##
## $cluster.size
## [1] 3 7 7 3
##
## $min.cluster.size
## [1] 3
##
## $noisen
## [1] 0
##
## $diameter
## [1] 2.006881 2.655233 3.042582 3.149758
##
## $average.distance
## [1] 1.884657 1.510144 1.808694 2.414741
##
## $median.distance
## [1] 1.841812 1.226519 1.973881 2.894241
##
## $separation
## [1] 1.961110 1.622724 1.622724 1.786151
##
## $average.toother
## [1] 3.567784 3.034966 2.797144 3.580547
##
## $separation.matrix
##          [,1]     [,2]     [,3]     [,4]
## [1,] 0.000000 3.555662 1.961110 3.647590
## [2,] 3.555662 0.000000 1.622724 1.984291
## [3,] 1.961110 1.622724 0.000000 1.786151
## [4,] 3.647590 1.984291 1.786151 0.000000
##
## $ave.between.matrix
##          [,1]     [,2]     [,3]     [,4]
## [1,] 0.000000 4.017357 2.753350 4.419129
## [2,] 4.017357 0.000000 2.507157 3.284129
## [3,] 2.753350 2.507157 0.000000 3.517574
## [4,] 4.419129 3.284129 3.517574 0.000000
##
## $average.between
## [1] 3.152419
##
## $average.within
## [1] 1.806503
##
## $n.between
## [1] 142
##
## $n.within
## [1] 48
##
## $max.diameter
## [1] 3.149758
##
## $min.separation
## [1] 1.622724
##
## $within.cluster.ss
## [1] 31.5904
##
## $clus.avg.silwidths
##         1         2         3         4
## 0.3152239 0.4024257 0.2528285 0.2370705
##
## $avg.silwidth
## [1] 0.3121831
##
## $g2
## NULL
##
## $g3
## NULL
##
## $pearsongamma
## [1] 0.5651189
##
## $dunn
## [1] 0.51519
##
## $dunn2
## [1] 1.038272
##
## $entropy
## [1] 1.304011
##
## $wb.ratio
## [1] 0.573053
##
## $ch
## [1] 9.10143
##
## $cwidegap
## [1] 1.841812 2.311732 1.833560 2.894241
##
## $widestgap
## [1] 2.894241
##
## $sindex
## [1] 1.622724
##
## $corrected.rand
## [1] 0.7443291
##
## $vi
## [1] 0.3054322
```

```
"comparative statistics of clusters- osobno dla dw?ch podzia??w na 2 i 3 klastry"
```

```
## [1] "comparative statistics of clusters- osobno dla dw?ch podzia??w na 2 i 3 klastry"
```

```
R3<-cluster.stats(dist.K,person.clust3)
R2<-cluster.stats(dist.K,person.clust2)
R3
```

```
## $n
## [1] 20
##
## $cluster.number
## [1] 4
##
## $cluster.size
## [1] 3 7 7 3
##
## $min.cluster.size
## [1] 3
##
## $noisen
## [1] 0
##
## $diameter
## [1] 2.006881 2.655233 3.042582 3.149758
##
## $average.distance
## [1] 1.884657 1.510144 1.808694 2.414741
##
## $median.distance
## [1] 1.841812 1.226519 1.973881 2.894241
##
## $separation
## [1] 1.961110 1.622724 1.622724 1.786151
##
## $average.toother
## [1] 3.567784 3.034966 2.797144 3.580547
##
## $separation.matrix
##          [,1]     [,2]     [,3]     [,4]
## [1,] 0.000000 3.555662 1.961110 3.647590
## [2,] 3.555662 0.000000 1.622724 1.984291
## [3,] 1.961110 1.622724 0.000000 1.786151
## [4,] 3.647590 1.984291 1.786151 0.000000
##
## $ave.between.matrix
##          [,1]     [,2]     [,3]     [,4]
## [1,] 0.000000 4.017357 2.753350 4.419129
## [2,] 4.017357 0.000000 2.507157 3.284129
## [3,] 2.753350 2.507157 0.000000 3.517574
## [4,] 4.419129 3.284129 3.517574 0.000000
##
## $average.between
## [1] 3.152419
##
## $average.within
## [1] 1.806503
##
## $n.between
## [1] 142
##
## $n.within
## [1] 48
##
## $max.diameter
## [1] 3.149758
##
## $min.separation
## [1] 1.622724
##
## $within.cluster.ss
## [1] 31.5904
##
## $clus.avg.silwidths
##         1         2         3         4
## 0.3152239 0.4024257 0.2528285 0.2370705
##
## $avg.silwidth
## [1] 0.3121831
##
## $g2
## NULL
##
## $g3
## NULL
##
## $pearsongamma
## [1] 0.5651189
##
## $dunn
## [1] 0.51519
##
## $dunn2
## [1] 1.038272
##
## $entropy
## [1] 1.304011
##
## $wb.ratio
## [1] 0.573053
##
## $ch
## [1] 9.10143
##
## $cwidegap
## [1] 1.841812 2.311732 1.833560 2.894241
##
## $widestgap
## [1] 2.894241
##
## $sindex
## [1] 1.622724
##
## $corrected.rand
## NULL
##
## $vi
## NULL
```

```r
R2
```

```
## $n
## [1] 20
##
## $cluster.number
## [1] 3
##
## $cluster.size
## [1] 10  7  3
##
## $min.cluster.size
## [1] 3
##
## $noisen
## [1] 0
##
## $diameter
## [1] 3.774667 2.655233 3.149758
##
## $average.distance
## [1] 2.254598 1.510144 2.414741
##
## $median.distance
## [1] 2.268880 1.226519 2.894241
##
## $separation
## [1] 1.622724 1.622724 1.786151
##
```

```
## $average.toother
## [1] 3.208564 3.034966 3.580547
## 
## $separation.matrix
##          [,1]     [,2]     [,3]
## [1,] 0.000000 1.622724 1.786151
## [2,] 1.622724 0.000000 1.984291
## [3,] 1.786151 1.984291 0.000000
## 
## $ave.between.matrix
##          [,1]     [,2]     [,3]
## [1,] 0.000000 2.960217 3.788040
## [2,] 2.960217 0.000000 3.284129
## [3,] 3.788040 3.284129 0.000000
## 
## $average.between
## [1] 3.221678
## 
## $average.within
## [1] 2.01806
## 
## $n.between
## [1] 121
## 
## $n.within
## [1] 69
## 
## $max.diameter
## [1] 3.774667
## 
## $min.separation
## [1] 1.622724
## 
## $within.cluster.ss
## [1] 41.85139
## 
## $clus.avg.silwidths
##         1         2         3 
## 0.2130971 0.4925152 0.2395034 
## 
## $avg.silwidth
## [1] 0.3148544
## 
## $g2
## NULL
## 
## $g3
## NULL
## 
## $pearsongamma
## [1] 0.5184101
## 
## $dunn
## [1] 0.4298985
## 
## $dunn2
## [1] 1.225894
## 
## $entropy
## [1] 0.9985793
## 
## $wb.ratio
## [1] 0.6264003
## 
## $ch
## [1] 8.865015
## 
## $cwidegap
## [1] 1.961110 2.311732 2.894241
## 
## $widestgap
## [1] 2.894241
## 
## $sindex
## [1] 1.622724
## 
## $corrected.rand
## NULL
## 
## $vi
## NULL
```

```
"the size of clusters created - ilosci elementow w klastrach"
```

```
## [1] "the size of clusters created - ilosci elementow w klastrach"
```

```
R3$cluster.size
```

```
## [1] 3 7 7 3
```

```
R2$cluster.size
```

```
## [1] 10  7  3
```

```
"intra-group diversity - zroznicowanie odleglosci
wewnatrz klastrow czym mniejsze tym elementy w klastrach blizsze"
```

```
## [1] "intra-group diversity - zroznicowanie odleglosci\nwewnatrz klastrow czym mniejsze tym elementy w klastrach blizsze"
```

```
R3$average.distance
```

```
## [1] 1.884657 1.510144 1.808694 2.414741
```

```
R2$average.distance
```

```
## [1] 2.254598 1.510144 2.414741
```

```
"separation matrix, the smallest distances between
observations belonging to different groups in cluster"
```

```
## [1] "separation matrix, the smallest distances between \nobservations belonging to different groups in cluster"
```

```
R3$separation.matrix
```

```
##          [,1]     [,2]     [,3]     [,4]
## [1,] 0.000000 3.555662 1.961110 3.647590
## [2,] 3.555662 0.000000 1.622724 1.984291
## [3,] 1.961110 1.622724 0.000000 1.786151
## [4,] 3.647590 1.984291 1.786151 0.000000
```

```
R2$separation.matrix
```

```
##          [,1]     [,2]     [,3]
## [1,] 0.000000 1.622724 1.786151
## [2,] 1.622724 0.000000 1.984291
## [3,] 1.786151 1.984291 0.000000
```

```
"the average distances between group of clusters"
```

```
## [1] "the average distances between group of clusters"
```

```
R3$average.between
```

```
## [1] 3.152419
```

```
R2$average.between
```

```
## [1] 3.221678
```

```
"the average distances within group of clusters"
```

```
## [1] "the average distances within group of clusters"
```

```
R3$average.within
```

```
## [1] 1.806503
```

```
R2$average.within
```

```
## [1] 2.01806
```

```
"badanie kt?ry z podzia??w jest lepszy"
```

```
## [1] "badanie kt?ry z podzia??w jest lepszy"
```

```
"the ratio between average group and outside group differences,
the lower the better"
```

```
## [1] "the ratio between average group and outside group differences,\nthe lower the better"
```

```
R11<-R3$wb.ratio
R12<-R2$wb.ratio
R11
```

```
## [1] 0.573053
```

```
R12
```

```
## [1] 0.6264003
```

```
ifelse(R11<=R12,"better is first clustering","better is second clastering")
```

```
## [1] "better is first clustering"
```

```
"Gamma coefficient, the higher the better"
```

```
## [1] "Gamma coefficient, the higher the better"
```

```
R13<-R3$pearsongamma
R14<-R2$pearsongamma
R13
```

```
## [1] 0.5651189
```

```
R14
```

```
## [1] 0.5184101
```

```
ifelse(R13>=R14,"better is first clustering","better is second clastering")
```

```
## [1] "better is first clustering"
```

```
"Dunn's Index, the higher the better"
```

```
## [1] "Dunn's Index, the higher the better"
```

```
R15<-R3$dunn2
R16<-R2$dunn2
ifelse(R15>=R16,"better is first clustering","better is second clastering")
```

```
## [1] "better is second clastering"
```

```
" 2. metoda clusteringu - niehierarchiczna - k-?rednich"
```

```
## [1] " 2. metoda clusteringu - niehierarchiczna - k-?rednich"
```

```
# k-means method
cluster3<-kmeans(K1[,2:k],4)
cluster2<-kmeans(K1[,2:k],3)
# summary of the cluster3
cluster3
```

```
## K-means clustering with 4 clusters of sizes 7, 6, 3, 4
##
## Cluster means:
##         Sex Marital.status        Age  Education     Income  Occupation
## 1 0.0000000      0.0000000  0.1745402 -0.6981175  0.2139376 -0.06688869
## 2 1.0000000      1.0000000 -0.9678155 -0.1809934 -0.6419063 -0.10404908
## 3 0.3333333      0.6666667 -0.2699816  0.4223180  1.6650851  1.71680980
## 4 0.0000000      0.2500000  1.3487641  1.1764573 -0.6603453 -1.01447852
##
## Clustering vector:
##  [1] 4 2 4 1 1 1 1 3 4 3 2 2 2 4 2 2 1 1 3 1
##
## Within cluster sum of squares by cluster:
## [1] 11.390525  3.766841  6.579382  7.428966
##  (between_SS / total_SS =  65.9 %)
##
## Available components:
##
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```
"przypisanie do klastrow elementoW"
```

```
## [1] "przypisanie do klastrow elementoW"
```

```
cluster3$cluster
```

```
##  [1] 4 2 4 1 1 1 1 3 4 3 2 2 2 4 2 2 1 1 3 1
```

```
"srednie wartosci zmiennych w klastrach"
```

```
## [1] "srednie wartosci zmiennych w klastrach"
```

```r
cluster3$centers
```

```
##         Sex Marital.status        Age  Education     Income  Occupation
## 1 0.0000000      0.0000000  0.1745402 -0.6981175  0.2139376 -0.06688869
## 2 1.0000000      1.0000000 -0.9678155 -0.1809934 -0.6419063 -0.10404908
## 3 0.3333333      0.6666667 -0.2699816  0.4223180  1.6650851  1.71680980
## 4 0.0000000      0.2500000  1.3487641  1.1764573 -0.6603453 -1.01447852
```

```r
"podstawowe informacje"
```

```
## [1] "podstawowe informacje"
```

```r
str(cluster3)
```

```
## List of 9
##  $ cluster     : int [1:20] 4 2 4 1 1 1 1 3 4 3 ...
##  $ centers     : num [1:4, 1:6] 0 1 0.333 0 0 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:4] "1" "2" "3" "4"
##   .. ..$ : chr [1:6] "Sex" "Marital.status" "Age" "Education" ...
##  $ totss       : num 85.5
##  $ withinss    : num [1:4] 11.39 3.77 6.58 7.43
##  $ tot.withinss: num 29.2
##  $ betweenss   : num 56.3
##  $ size        : int [1:4] 7 6 3 4
##  $ iter        : int 3
##  $ ifault      : int 0
##  - attr(*, "class")= chr "kmeans"
```

```r
#summary of the cluster2
cluster2
```

```
## K-means clustering with 3 clusters of sizes 8, 6, 6
## 
## Cluster means:
##     Sex Marital.status         Age  Education     Income Occupation
## 1 0.125      0.2500000 -0.02402379 -0.4072352  0.8470515  0.7413497
## 2 0.000      0.1666667  0.99984721  0.7239737 -0.4874957 -0.8844172
## 3 1.000      1.0000000 -0.96781549 -0.1809934 -0.6419063 -0.1040491
## 
## Clustering vector:
##  [1] 2 3 2 1 1 2 1 1 2 1 3 3 3 2 3 3 2 1 1 1
## 
## Within cluster sum of squares by cluster:
## [1] 26.276287 14.081259  3.766841
##  (between_SS / total_SS =  48.4 %)
## 
## Available components:
## 
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```r
"przypisanie do klastrow elementoW"
```

```
## [1] "przypisanie do klastrow elementoW"
```

```r
cluster2$cluster
```

```
##  [1] 2 3 2 1 1 2 1 1 2 1 3 3 3 2 3 3 2 1 1 1
```

```r
"srednie wartosci zmiennych w klastrach"
```

```
## [1] "srednie wartosci zmiennych w klastrach"
```

```r
cluster3$centers
```

```
##         Sex Marital.status        Age  Education     Income  Occupation
## 1 0.0000000      0.0000000  0.1745402 -0.6981175  0.2139376 -0.06688869
## 2 1.0000000      1.0000000 -0.9678155 -0.1809934 -0.6419063 -0.10404908
## 3 0.3333333      0.6666667 -0.2699816  0.4223180  1.6650851  1.71680980
## 4 0.0000000      0.2500000  1.3487641  1.1764573 -0.6603453 -1.01447852
```

```r
"podstawowe informacje"
```

```
## [1] "podstawowe informacje"
```

```r
str(cluster2)
```

```
## List of 9
##  $ cluster     : int [1:20] 2 3 2 1 1 2 1 1 2 1 ...
##  $ centers     : num [1:3, 1:6] 0.125 0 1 0.25 0.167 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:3] "1" "2" "3"
##   .. ..$ : chr [1:6] "Sex" "Marital.status" "Age" "Education" ...
##  $ totss       : num 85.5
##  $ withinss    : num [1:3] 26.28 14.08 3.77
##  $ tot.withinss: num 44.1
##  $ betweenss   : num 41.4
##  $ size        : int [1:3] 8 6 6
##  $ iter        : int 2
##  $ ifault      : int 0
##  - attr(*, "class")= chr "kmeans"
```

```r
"inna metoda jest zastosowanie nie k- srednich a centroidow"
```

```
## [1] "inna metoda jest zastosowanie nie k- srednich a centroidow"
```

```r
#Partitioning Around Medoids
# for 4 groups
klaster3<-pam(K1[,2:k],4)
klaster3
```

```
## Medoids:
##      ID Sex Marital.status        Age  Education      Income Occupation
## [1,] 14   0              0  1.4002437  1.6289408 -1.24765910 -1.4046626
## [2,] 11   1              1 -1.0021352 -0.1809934 -0.78685287  0.1560736
## [3,]  4   0              0  0.3706527 -0.1809934  0.73579859  0.1560736
## [4,] 20   0              0 -0.5902988 -1.9909277  0.08644672  0.1560736
## Clustering vector:
##  [1] 1 2 1 3 3 3 3 3 1 3 2 2 2 1 2 2 3 4 3 4
## Objective function:
##    build     swap 
## 1.089425 1.089425 
## 
## Available components:
##  [1] "medoids"    "id.med"     "clustering" "objective"  "isolation" 
##  [6] "clusinfo"   "silinfo"    "diss"       "call"       "data"
```

```r
str(klaster3)
```

```
## List of 10
##  $ medoids   : num [1:4, 1:6] 0 1 0 0 0 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : NULL
##   .. ..$ : chr [1:6] "Sex" "Marital.status" "Age" "Education" ...
##  $ id.med    : int [1:4] 14 11 4 20
##  $ clustering: int [1:20] 1 2 1 3 3 3 3 3 1 3 ...
##  $ objective : Named num [1:2] 1.09 1.09
##   ..- attr(*, "names")= chr [1:2] "build" "swap"
##  $ isolation : Factor w/ 3 levels "no","L","L*": 1 1 1 3
##   ..- attr(*, "names")= chr [1:4] "1" "2" "3" "4"
##  $ clusinfo  : num [1:4, 1:5] 4 6 8 2 1.96 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : NULL
##   .. ..$ : chr [1:5] "size" "max_diss" "av_diss" "diameter" ...
##  $ silinfo   :List of 3
##   ..$ widths         : num [1:20, 1:3] 1 1 1 1 2 2 2 2 2 2 ...
##   .. ..- attr(*, "dimnames")=List of 2
##   .. .. ..$ : chr [1:20] "14" "9" "1" "3" ...
##   .. .. ..$ : chr [1:3] "cluster" "neighbor" "sil_width"
##   ..$ clus.avg.widths: num [1:4] 0.291 0.599 0.133 0.802
##   ..$ avg.width      : num 0.371
##  $ diss      : 'dissimilarity' num [1:190] 3.9 2.82 2.61 2.13 3.28 ...
##   ..- attr(*, "Size")= int 20
##   ..- attr(*, "Metric")= chr "euclidean"
##  $ call      : language pam(x = K1[, 2:k], k = 4)
##  $ data      : num [1:20, 1:6] 0 1 0 0 0 0 0 0 0 0 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : NULL
##   .. ..$ : chr [1:6] "Sex" "Marital.status" "Age" "Education" ...
##  - attr(*, "class")= chr [1:2] "pam" "partition"
```

```
#for 3 groups
klaster2<-pam(K1[,2:k],3)
str(klaster2)
```

```
## List of 10
##  $ medoids   : num [1:3, 1:6] 0 1 0 0 1 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : NULL
##   .. ..$ : chr [1:6] "Sex" "Marital.status" "Age" "Education" ...
##  $ id.med    : int [1:3] 14 11 4
##  $ clustering: int [1:20] 1 2 1 3 3 3 3 3 1 3 ...
##  $ objective : Named num [1:2] 1.29 1.29
##   ..- attr(*, "names")= chr [1:2] "build" "swap"
##  $ isolation : Factor w/ 3 levels "no","L","L*": 1 1 1
##   ..- attr(*, "names")= chr [1:3] "1" "2" "3"
##  $ clusinfo  : num [1:3, 1:5] 4 6 10 1.96 1.7 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : NULL
##   .. ..$ : chr [1:5] "size" "max_diss" "av_diss" "diameter" ...
##  $ silinfo   :List of 3
##   ..$ widths         : num [1:20, 1:3] 1 1 1 1 2 2 2 2 2 2 ...
##   .. ..- attr(*, "dimnames")=List of 2
##   .. .. ..$ : chr [1:20] "14" "9" "1" "3" ...
##   .. .. ..$ : chr [1:3] "cluster" "neighbor" "sil_width"
##   ..$ clus.avg.widths: num [1:3] 0.3207 0.6222 0.0995
##   ..$ avg.width      : num 0.301
##  $ diss      : 'dissimilarity' num [1:190] 3.9 2.82 2.61 2.13 3.28 ...
##   ..- attr(*, "Size")= int 20
##   ..- attr(*, "Metric")= chr "euclidean"
##  $ call      : language pam(x = K1[, 2:k], k = 3)
##  $ data      : num [1:20, 1:6] 0 1 0 0 0 0 0 0 0 0 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : NULL
##   .. ..$ : chr [1:6] "Sex" "Marital.status" "Age" "Education" ...
##  - attr(*, "class")= chr [1:2] "pam" "partition"
```

```
"podsumowania"
```

```
## [1] "podsumowania"
```

```
S3<-summary(klaster3)
S2<-summary(klaster2)

"przeglad podsumowania na przykladzie S3"
```

```
## [1] "przeglad podsumowania na przykladzie S3"
```

```
"wspolrzedne punktow bedacych centroidami"
```

```
## [1] "wspolrzedne punktow bedacych centroidami"
```

```
S3$medoids
```

```
##      Sex Marital.status        Age  Education      Income Occupation
## [1,]   0              0  1.4002437  1.6289408 -1.24765910 -1.4046626
## [2,]   1              1 -1.0021352 -0.1809934 -0.78685287  0.1560736
## [3,]   0              0  0.3706527 -0.1809934  0.73579859  0.1560736
## [4,]   0              0 -0.5902988 -1.9909277  0.08644672  0.1560736
```

```
"identyfikacja median"
```

```
## [1] "identyfikacja median"
```

```
S3$id.med
```

```
## [1] 14 11  4 20
```

```
"przypisanie elementoW do klastrow"
```

```
## [1] "przypisanie elementoW do klastrow"
```

```
S3$clustering
```

```
##  [1] 1 2 1 3 3 3 3 3 1 3 2 2 2 1 2 2 3 4 3 4
```

```
"parametry klastrow"
```

```
## [1] "parametry klastrow"
```

```
S3$objective
```

```
##    build     swap
## 1.089425 1.089425
```

```
"informacje o klastrach"
```

```
## [1] "informacje o klastrach"
```

```
S3$clusinfo
```

```
##      size  max_diss   av_diss  diameter separation
## [1,]    4 1.9611102 1.4020503 2.8232106   1.651120
## [2,]    6 1.7031317 0.6105218 2.2739039   1.833560
## [3,]    8 3.5318982 1.5003374 4.8360401   1.651120
## [4,]    2 0.5144723 0.2572362 0.5144723   2.149639
```

```
"informacje o przyporzadkowaniu i sasiadach elementoW"
```

```
## [1] "informacje o przyporzadkowaniu i sasiadach elementoW"
```

```
S3$silinfo
```

```
## $widths
##    cluster neighbor   sil_width
## 14       1        2  0.49033625
## 9        1        3  0.34748554
## 1        1        3  0.26056101
## 3        1        2  0.06667715
## 11       2        4  0.71126082
## 13       2        4  0.69483681
## 15       2        4  0.67623503
## 12       2        4  0.65926964
## 2        2        4  0.47743428
## 16       2        1  0.37373606
## 7        3        4  0.30718090
## 5        3        1  0.27863185
## 4        3        4  0.22879720
## 19       3        2  0.17781993
## 8        3        4  0.12692733
## 17       3        1  0.10792617
## 10       3        2  0.01327891
## 6        3        2 -0.17400188
## 18       4        2  0.80488361
## 20       4        2  0.79952321
## 
## $clus.avg.widths
## [1] 0.2912650 0.5987954 0.1333200 0.8022034
## 
## $avg.width
## [1] 0.37144
```

```
"informacje o odleglosciach"
```

```
## [1] "informacje o odleglosciach"
```

```
S3$diss
```

```
## Dissimilarities :
##            1         2         3         4         5         6         7
## 2  3.9004084                                                            
## 3  2.8232106 3.1745330                                                  
## 4  2.6147414 2.1780897 2.5418692                                        
## 5  2.1318773 2.5552673 2.1436681 0.7728149                              
## 6  3.2822723 2.2918470 1.6511203 1.8228349 1.9931308                    
## 7  2.1884081 2.5586502 2.2688800 0.6586244 0.1801235 2.0103184          
## 8  3.6475904 2.5102425 4.1250202 1.7861514 2.2628044 3.3360050 2.1829066
## 9  2.0068812 3.7253373 2.6871023 2.8548575 2.6489816 2.7362794 2.6509041
## 10 3.9156443 1.9842907 4.1318501 2.1915772 2.6005018 3.3899179 2.5636630
## 11 3.7066924 1.0414530 2.7139731 2.4905851 2.5791214 2.3828047 2.6527233
## 12 3.7406103 0.5759149 2.8702982 2.2811376 2.4959635 2.2758084 2.5394143
## 13 3.8684815 1.0156319 2.8445399 2.6066813 2.7341375 2.4481897 2.8033348
## 14 1.8418123 4.0878362 1.9611102 3.2719904 2.8311197 2.8306579 2.9268270
## 15 3.5662136 1.2265193 2.5734683 2.4713277 2.4867603 2.3856986 2.5724451
## 16 3.7746674 2.2739039 1.8335597 3.0425820 2.9415109 1.9738807 3.0350928
## 17 2.0493269 2.6264283 1.8111741 1.2387984 0.5666505 2.0443242 0.7467740
## 18 4.6946203 2.3117317 3.3370864 2.2971067 2.6477717 2.5050245 2.6400923
## 19 4.2312783 3.8975089 5.7021903 3.5318982 3.9394467 4.8360401 3.8170665
## 20 4.4093111 2.3831212 3.0047723 2.1496390 2.3595138 2.4056522 2.3744000
##            8         9        10        11        12        13        14
## 2                                                                       
## 3                                                                       
## 4                                                                       
## 5                                                                       
## 6                                                                       
## 7                                                                       
## 8                                                                       
## 9  4.2700235                                                            
## 10 1.2002245 4.2964727                                                  
## 11 3.0215399 3.7283519 2.4553449                                        
## 12 2.7467597 3.6738161 2.1909971 0.4666541                              
## 13 3.0713505 3.8664899 2.4777308 0.2059854 0.4765199                    
## 14 4.7215359 1.8052786 4.8040057 3.7007467 3.8299457 3.8382537          
## 15 3.0756986 3.6363691 2.5364764 0.2459069 0.6561953 0.4348782 3.5556616
## 16 4.2696864 3.2498485 3.8920468 1.7031317 1.9054800 1.7737118 2.9950117
## 17 2.5808671 2.7218027 2.7900271 2.4214920 2.4422730 2.5871270 2.5893557
## 18 2.6354328 4.7607940 2.6318606 2.5830467 2.4081236 2.5888468 4.8998360
## 19 2.8942410 4.3815697 3.1497581 4.5968924 4.2554599 4.6553167 5.5040400
## 20 2.6801777 4.5615322 2.6977599 2.4916106 2.3824847 2.5319901 4.6131815
##           15        16        17        18        19
## 2                                                   
## 3                                                   
## 4                                                   
## 5                                                   
## 6                                                   
## 7                                                   
## 8                                                   
## 9                                                   
## 10                                                  
## 11                                                  
## 12                                                  
## 13                                                  
## 14                                                  
## 15                                                  
## 16 1.6227239                                        
## 17 2.2898468 2.7048360                              
## 18 2.6552335 3.2734923 2.7498368                    
## 19 4.6530126 5.5886778 4.3507043 5.0598362          
## 20 2.5166956 3.0915600 2.4018270 0.5144723 5.1330158
## 
## Metric :  euclidean 
## Number of objects : 20
```

```
"tabela zbiorcza"
```

```
## [1] "tabela zbiorcza"
```

```
D1<-data.frame(cluster3$cluster)
D2<-data.frame(cluster2$cluster)
D3<-data.frame(klaster3$clustering)
D4<-data.frame(klaster2$clustering)
D<-cbind(K1[1],D1,D2,D3,D4,B[2],B1[2])
colnames(D)<-c("Id","srednia na 4","srednia na 3","med na 4", "med na 3", "dend4","dend3")
D
```

```
##    Id srednia na 4 srednia na 3 med na 4 med na 3 dend4 dend3
## 1   1            4            2        1        1     1     1
## 2   2            2            3        2        2     2     2
## 3   3            4            2        1        1     3     1
## 4   4            1            1        3        3     3     1
## 5   5            1            1        3        3     3     1
## 6   6            1            2        3        3     3     1
## 7   7            1            1        3        3     3     1
## 8   8            3            1        3        3     4     3
## 9   9            4            2        1        1     1     1
## 10 10            3            1        3        3     4     3
## 11 11            2            3        2        2     2     2
## 12 12            2            3        2        2     2     2
## 13 13            2            3        2        2     2     2
## 14 14            4            2        1        1     1     1
## 15 15            2            3        2        2     2     2
## 16 16            2            3        2        2     3     1
## 17 17            1            2        3        3     3     1
## 18 18            1            1        4        3     2     2
## 19 19            3            1        3        3     4     3
## 20 20            1            1        4        3     2     2
```

```
"wybrano podzial k-means na 4 klastry"
```

```
## [1] "wybrano podzial k-means na 4 klastry"
```

```
cluster3
```

```
## K-means clustering with 4 clusters of sizes 7, 6, 3, 4
## 
## Cluster means:
##         Sex Marital.status        Age  Education     Income  Occupation
## 1 0.0000000      0.0000000  0.1745402 -0.6981175  0.2139376 -0.06688869
## 2 1.0000000      1.0000000 -0.9678155 -0.1809934 -0.6419063 -0.10404908
## 3 0.3333333      0.6666667 -0.2699816  0.4223180  1.6650851  1.71680980
## 4 0.0000000      0.2500000  1.3487641  1.1764573 -0.6603453 -1.01447852
## 
## Clustering vector:
##  [1] 4 2 4 1 1 1 1 3 4 3 2 2 2 4 2 2 1 1 3 1
## 
## Within cluster sum of squares by cluster:
## [1] 11.390525  3.766841  6.579382  7.428966
##  (between_SS / total_SS =  65.9 %)
## 
## Available components:
## 
## [1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss"
## [6] "betweenss"    "size"         "iter"         "ifault"
```

```
"stworzyc ramke z naszymi danymi"
```

```
## [1] "stworzyc ramke z naszymi danymi"
```

```
G<-data.frame(K1,D1,D2)
G
```

```
##    id Sex Marital.status        Age  Education      Income Occupation
## 1   1   0              0  1.8807195  1.6289408 -0.39588552  0.1560736
## 2   2   1              1 -1.2080534 -0.1809934  0.23403985  0.1560736
## 3   3   0              0  0.6452103 -0.1809934 -1.25161680 -1.4046626
## 4   4   0              0  0.3706527 -0.1809934  0.73579859  0.1560736
## 5   5   0              0  0.9197679 -0.1809934  0.19200139  0.1560736
## 6   6   0              0 -0.3157412 -0.1809934  0.09105599 -1.4046626
## 7   7   0              0  0.9197679 -0.1809934  0.37212486  0.1560736
## 8   8   0              0 -0.3157412 -0.1809934  1.26806055  1.7168098
## 9   9   0              1  1.4688831  1.6289408  0.25378007 -1.4046626
## 10 10   0              1 -0.7962170 -0.1809934  0.81015020  1.7168098
## 11 11   1              1 -1.0021352 -0.1809934 -0.78685287  0.1560736
## 12 12   1              1 -1.0707746 -0.1809934 -0.32527442  0.1560736
## 13 13   1              1 -1.2080534 -0.1809934 -0.78159203  0.1560736
## 14 14   0              0  1.4002437  1.6289408 -1.24765910 -1.4046626
## 15 15   1              1 -0.7962170 -0.1809934 -0.92126977  0.1560736
## 16 16   1              1 -0.5216594 -0.1809934 -1.27048825 -1.4046626
## 17 17   0              0  0.9197679 -0.1809934 -0.37464910  0.1560736
## 18 18   0              0 -1.0021352 -1.9909277  0.39478509  0.1560736
## 19 19   1              1  0.3020133  1.6289408  2.91704459  1.7168098
## 20 20   0              0 -0.5902988 -1.9909277  0.08644672  0.1560736
##    Settlement.size cluster3.cluster cluster2.cluster
## 1        1.4396969                4                2
## 2        1.4396969                2                3
## 3       -0.9597980                4                2
## 4        0.2399495                1                1
## 5        0.2399495                1                1
## 6       -0.9597980                1                2
## 7        0.2399495                1                1
## 8        0.2399495                3                1
## 9       -0.9597980                4                2
## 10      -0.9597980                3                1
## 11      -0.9597980                2                3
## 12      -0.9597980                2                3
## 13       1.4396969                2                3
## 14      -0.9597980                4                2
## 15       0.2399495                2                3
## 16      -0.9597980                2                3
## 17      -0.9597980                1                2
## 18       1.4396969                1                1
## 19       1.4396969                3                1
## 20       0.2399495                1                1
```

```
write.table(G, file="lab2result.csv",append=TRUE,sep=";",row.names=TRUE,col.names=TRUE,dec = ",")
```

```
## Warning in write.table(G, file = "lab2result.csv", append = TRUE, sep = ";", :
## dołączanie nazw kolumn do pliku
```

```
ANOV<-anova(aov(G$Income~G$cluster3.cluster,G))
ANOV
```

```
## Analysis of Variance Table
## 
## Response: G$Income
##                    Df  Sum Sq Mean Sq F value Pr(>F)
## G$cluster3.cluster  1  0.1264 0.12645  0.1206 0.7324
## Residuals          18 18.8736 1.04853
```

```
ifelse(ANOV$`Pr(>F)`<0.05,"zmienna istotnie roznicuje klaster","zmienna nieistotnie roznicuje klaster")
```

```
## [1] "zmienna nieistotnie roznicuje klaster"
## [2] NA
```

```
"tabele porownujace clustering roznymi metodami"
```

```
## [1] "tabele porownujace clustering roznymi metodami"
```

```
D
```

```
##    Id srednia na 4 srednia na 3 med na 4 med na 3 dend4 dend3
## 1   1            4            2        1        1     1     1
## 2   2            2            3        2        2     2     2
## 3   3            4            2        1        1     3     1
## 4   4            1            1        3        3     3     1
## 5   5            1            1        3        3     3     1
## 6   6            1            2        3        3     3     1
## 7   7            1            1        3        3     3     1
## 8   8            3            1        3        3     4     3
## 9   9            4            2        1        1     1     1
## 10 10            3            1        3        3     4     3
## 11 11            2            3        2        2     2     2
## 12 12            2            3        2        2     2     2
## 13 13            2            3        2        2     2     2
## 14 14            4            2        1        1     1     1
## 15 15            2            3        2        2     2     2
## 16 16            2            3        2        2     3     1
## 17 17            1            2        3        3     3     1
## 18 18            1            1        4        3     2     2
## 19 19            3            1        3        3     4     3
## 20 20            1            1        4        3     2     2
```

```
"porownanie metody hierachicznej na 4 klastry z niehierarchiczna centr"
```

```
## [1] "porownanie metody hierachicznej na 4 klastry z niehierarchiczna centr"
```

```
table(D$`med na 4`,D$dend3)
```

```
## 
##     1 2 3
##   1 4 0 0
##   2 1 5 0
##   3 5 0 3
##   4 0 2 0
```

```
"porownanie metody hierachicznej na 4 klastry z niehierarchiczna k-M"
```

```
## [1] "porownanie metody hierachicznej na 4 klastry z niehierarchiczna k-M"
```

```
table(D$`srednia na 4`,D$dend3)
```

```
## 
##     1 2 3
##   1 5 2 0
##   2 1 5 0
##   3 0 0 3
##   4 4 0 0
```